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SECTION:P5

WEEK-4 C PROGRAMMING PROGRAMS

PROGRAM 1

CODE

```c
#include<stdio.h>

int main(){
    int n,num=1;
    printf("Enter the number of rows here: ");
    scanf("%d", &n);

    for (int i=1; i<=n; i++)
    {
        for (int j=1; j<=i; j++)
        {
            printf("%d",num);
            num++;
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\C WEEK-4> cd "c:\Users\Ch
Enter the number of rows here: 4
1
23
456
78910
PS C:\Users\Chethan S\Documents\C WEEK-4>
```

PROGRAM 2

CODE

```c
#include<stdio.h>

int main()
{

int pin=1234,input,try=2;



while(try!=-1)
{
    printf("Enter PIN: ");
    scanf("%d",&input);

    if(input==pin)
    {
        printf("Access Granted!\n");
        break;
    }
    else if(input!=pin && try>0)
    {
        printf("Wrong PIN!. Attempts left: %d\n",try);
        try--;
    }
    else if(input!=pin && try==0)
    {
        printf("Wrong PIN!. Attempts left: %d\n",try);
        printf("Card Blocked!!");
        break;
    }
    else
    {}
}

return 0;


}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\c week-4 submit> cd
Enter PIN: 1234
Access Granted!
PS C:\Users\Chethan S\Documents\c week-4 submit> cd
Enter PIN: 2345
Wrong PIN!. Attempts left: 2
Enter PIN: 6786
Wrong PIN!. Attempts left: 1
Enter PIN: 3456
Wrong PIN!. Attempts left: 0
Card Blocked!!
PS C:\Users\Chethan S\Documents\c week-4 submit>
```

PROGRAM 3

CODE

```c
#include<stdio.h>


int main()
{
int num,limit,threshold,sum=0,count=0,max=0;

printf("Enter the base number: ");
scanf("%d",&num);

printf("Enter  the limit: ");
scanf("%d",&limit);

printf("Enter the threshold value : ");
scanf("%d",&threshold);

for(int i=1;i<=limit;i++)
{
printf("%d X %d = %d\n",num,i,num*i);
sum+=num*i;

//*********************************************************************
    {if(threshold<num*i)
        {count++;}
    else
    {}
    }

//*********************************************************************
    {if(max<num*i)
        {max=num*i;}
    }

}
printf("============Summary===============\n");
printf("Total Sum = %d\n",sum);
printf("Results greater than %d = %d\n",threshold,count);
printf("Max result = %d",max);
return 0;



}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\c week-4 submit> cd "c:
Enter the base number: 45
Enter  the limit: 20
Enter the threshold value : 405
45 X 1 = 45
45 X 2 = 90
45 X 3 = 135
45 X 4 = 180
45 X 5 = 225
45 X 6 = 270
45 X 7 = 315
45 X 8 = 360
45 X 9 = 405
45 X 10 = 450
45 X 11 = 495
45 X 12 = 540
45 X 13 = 585
45 X 14 = 630
45 X 15 = 675
45 X 16 = 720
45 X 17 = 765
45 X 18 = 810
45 X 19 = 855
45 X 20 = 900
============Summary===============
Total Sum = 9450
Results greater than 405 = 11
Max result = 900
PS C:\Users\Chethan S\Documents\c week-4 submit>
```

PROGRAM 4

CODE

```c
#include<stdio.h>

int main()
{
    float p,r,E,outstanding,total_interest=0,total_penalty=0,payment;
    int N,month=0;

    printf("Enter the principal amount: ");
    scanf("%f",&p);

    printf("Enter the monthly interest: ");
    scanf("%f",&r);

    printf("Enter the emi: ");
    scanf("%f",&E);

    printf("Enter the maximum months: ");
    scanf("%d",&N);

    outstanding=p;

    while(outstanding>0 && month<N)
    {
        month++;
        float interest=outstanding*r/100;
        total_interest+=interest;

        payment=E;

        if(payment<interest)
        {
            total_penalty+=200;
            payment-=200;
        }

        outstanding=outstanding+interest-payment;

    if(outstanding > 0)
        printf("Month %d outstanding: %.2f\n", month, outstanding);
    else
        printf("Month %d outstanding: 0.00\n", month);
    }

    printf("Total interest: %.2f\n",total_interest);
    printf("Total penalty: %.2f\n",total_penalty);

    if(outstanding<=0)
        printf("Loan status: Closed\n");
    else
        printf("Remaining Principal: %.2f\n",outstanding);
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\c week-4 submit> cd "c
Enter the principal amount: 10000
Enter the monthly interest: 2
Enter the EMI: 2500
Enter the maximum months: 6
Month 1 outstanding: 7700.00
Month 2 outstanding: 5354.00
Month 3 outstanding: 2961.08
Month 4 outstanding: 520.30
Month 5 outstanding: 0.00

Total interest paid: 530.71
Total penalty: 0.00
Months processed: 5
Loan status: Closed
PS C:\Users\Chethan S\Documents\c week-4 submit>
```

PROGRAM 5

CODE

```c
{
    int choice=1;
    float hours,rate,bill;
    int nov=0;
    float revenue=0;
    float max_bill=0,min_bill=0;

    while(choice!=0)
    {
        printf("Enter vehicle type (1=Bike,2=Car,3=SUV,4=Truck,0=Stop): ");
        scanf("%d",&choice);

        if(choice==0)
            break;

        printf("Enter hours parked: ");
        scanf("%f",&hours);

        if(choice==1)
            rate=20;
        else if(choice==2)
            rate=40;
        else if(choice==3)
            rate=60;
        else if(choice==4)
            rate=80;
        else
            {}
        bill=rate*hours;
        if(hours>6)
            bill=bill-(bill*0.15);
        nov++;
        if(nov%10==0)
            bill=bill+50;

        printf("Vehicle %d bill: %.2f\n",nov,bill);
        revenue+=bill;
        if(nov==1)
        {max_bill=bill;
            min_bill=bill;
        }
        else
        {
            if(bill>max_bill)
                max_bill=bill;

            if(bill<min_bill)
                min_bill=bill;
        }
    }

    printf("Total vehicles: %d\n",nov);
    printf("Total revenue: %.2f\n",revenue);
    printf("Maximum bill: %.2f\n",max_bill);
    printf("Minimum bill: %.2f\n",min_bill);
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\c week-4 submit> cd "c:\Users\C
Enter vehicle type (1=Bike,2=Car,3=SUV,4=Truck,0=Stop): 2
Enter hours parked: 4
Vehicle 1 bill: 160.00
Enter vehicle type (1=Bike,2=Car,3=SUV,4=Truck,0=Stop): 4
Enter hours parked: 5
Vehicle 2 bill: 400.00
Enter vehicle type (1=Bike,2=Car,3=SUV,4=Truck,0=Stop): 3
Enter hours parked: 8
Vehicle 3 bill: 408.00
Enter vehicle type (1=Bike,2=Car,3=SUV,4=Truck,0=Stop): 4
Enter hours parked: 12
Vehicle 4 bill: 816.00
Enter vehicle type (1=Bike,2=Car,3=SUV,4=Truck,0=Stop): 0
Total vehicles: 4
Total revenue: 1784.00
Maximum bill: 816.00
Minimum bill: 160.00
PS C:\Users\Chethan S\Documents\c week-4 submit>
```